

NIFTY 50

MARKET PERFORMANCE

ANALYTICS REPORT

A 16-Year Historical Analysis of Price Trends, Volatility & Trading Behaviour (2010–2025)

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1. Executive Summary

This report presents a comprehensive 16-year analysis of NIFTY 50 historical market data, covering 3,929 trading days from January 4, 2010 through December 31, 2025. The analysis examines price trends, trading volume behaviour, volatility, and day-to-day return patterns to convert raw market data into actionable business insight for investors and risk managers.

The headline finding is a striking divergence between price-level performance and trading-session performance. While the NIFTY 50 closing price climbed from roughly ₹5,200 to ₹26,216 — more than a 5x increase — the average open-to-close (intraday session) return across the full period is actually negative, at -0.06% per day, with Loss Days (53.0%) outnumbering Profit Days (47.0%). Essentially all of the index's long-term appreciation has been driven by overnight/opening gaps rather than gains accrued during the trading session itself — a critical structural insight for anyone building intraday strategies on this index.

Market risk is also highly episodic rather than evenly distributed: the 20-day rolling volatility measure spiked to more than 5x its historical baseline during the March 2020 COVID-19 crash, and 6 of the 10 largest single-day moves of either direction cluster within a single 4-month window (February–June 2020).

3,929 Total Trading Days	16 yrs Period Covered (2010–2025)	47.0% Bullish (Profit) Days	53.0% Bearish (Loss) Days
₹26,216 All-Time High Close	₹4,544 All-Time Low Close	+9.31% Best Single Day (13 Mar 2020)	−6.82% Worst Single Day (18 Mar 2020)

2. Business Context & Objectives

A financial research company sought to analyze historical NIFTY 50 market data to understand the overall market trend, price movement, trading volume behaviour, market volatility, and the best and worst trading periods — converting raw daily OHLCV data into business insight that supports investor decision-making.

Analytical Objectives

- Understand long-term historical market performance and growth trajectory
- Clean and preprocess 16 years of daily trading data
- Engineer business-relevant features: returns, ranges, moving averages, volatility, cumulative return
- Quantify volatility and identify the highest-risk periods in market history
- Identify seasonal and calendar-based patterns (month, quarter, weekday)
- Surface the relationship between price levels, trading volume, and realized returns

- Translate findings into actionable recommendations for investors and risk managers

3. Key Performance Indicators (KPIs)

The following KPIs were computed directly from the cleaned, feature-engineered dataset to establish a performance baseline.

KPI	Value
Total Trading Days	3,929
Date Range	04 Jan 2010 – 31 Dec 2025
Highest Closing Price	₹26,216.05
Lowest Closing Price	₹4,544.20
Average Closing Price	₹11,838.52
Highest Opening Price	₹26,325.80
Lowest Opening Price	₹4,623.15
Highest Daily Volume	1,811,000
Average Daily Volume	240,927
Highest Single-Day Gain	+9.31% (13 Mar 2020)
Highest Single-Day Loss	−6.82% (18 Mar 2020)
Average Daily Return (Open→Close)	−0.06%
Average Daily Trading Range	129.8 points
Average 20-Day Rolling Volatility	0.8
Profit (Bullish) Days	1,847 (47.0%)
Loss (Bearish) Days	2,081 (53.0%)

• Bearish Days Outnumber Bullish Days, Despite a Rising Market

53.0% of all trading sessions close lower than they open, versus 47.0% that close higher. Combined with a negative average open-to-close return (−0.06%), this confirms the index's long-term uptrend is not explained by typical trading-session performance — see Section 7 for the cumulative return anomaly that explains this.

• A More-Than-5x Price Increase Over 16 Years

The index moved from approximately ₹5,200 at the start of 2010 to ₹26,216 at the end of 2025, an annualized price-level growth rate of roughly 10–11% — a strong long-term return profile for buy-and-hold investors, even though the underlying intraday statistics above point to negative average session returns.

4. Price & Return Analysis

4.1 Closing Price Trend & Distribution

The closing price shows a clear long-term upward trajectory across the full 16-year window, punctuated by two notable episodes: a multi-year consolidation between 2010–2014, and a sharp, V-shaped correction in early 2020 (the COVID-19 crash) followed by an accelerated bull run through 2021–2025.

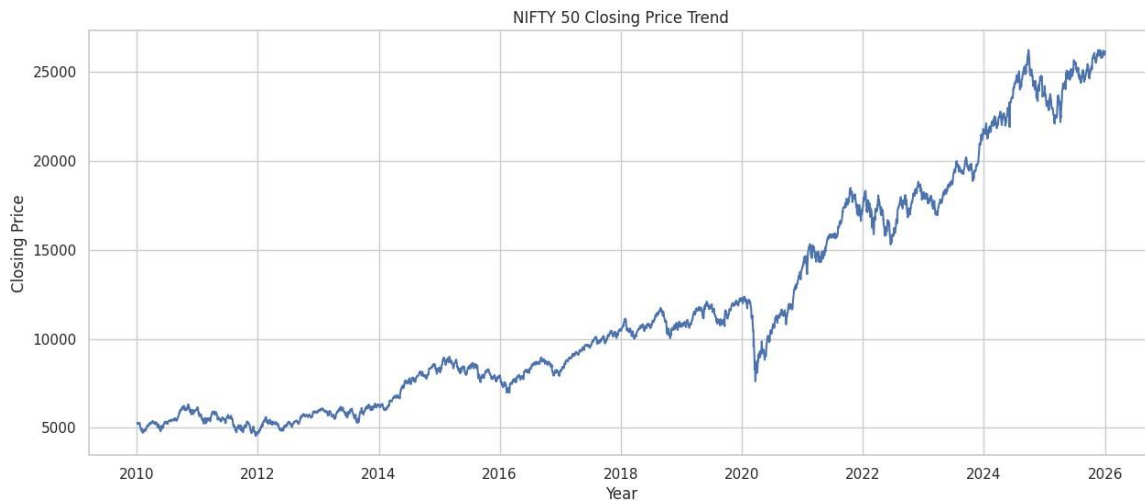


Figure 1: NIFTY 50 Closing Price Trend, 2010–2025

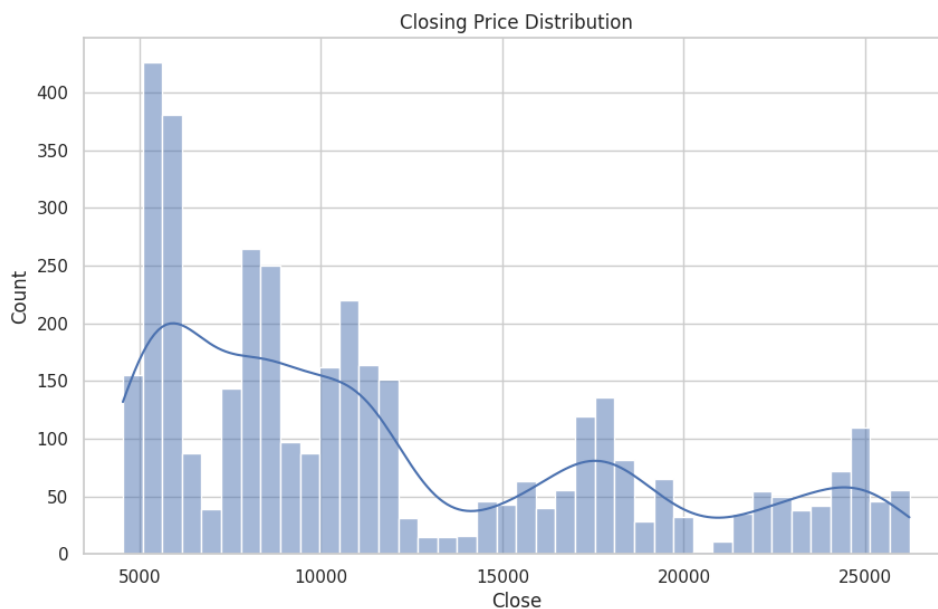


Figure 2: Closing Price Distribution — multi-modal, reflecting distinct multi-year price regimes (e.g., ₹5,000–6,000 in 2010–12, ₹17,000–18,000 in 2021–23, ₹24,000–26,000 in 2024–25)

The multi-modal shape of the distribution is a direct artifact of the index spending extended multi-year periods consolidating within a price band before breaking out to the next regime — most visibly the 2010–2014 band near ₹5,000–6,000 and the most recent 2024–2025 band near ₹24,000–26,000.

4.2 Daily Return Distribution

The daily open-to-close return distribution is tightly clustered around zero with a slight negative skew. Extreme single-day moves beyond $\pm 4\%$ are rare but not negligible, and — as Section 6 shows — almost all of them trace back to known market-stress events rather than being randomly distributed through time.

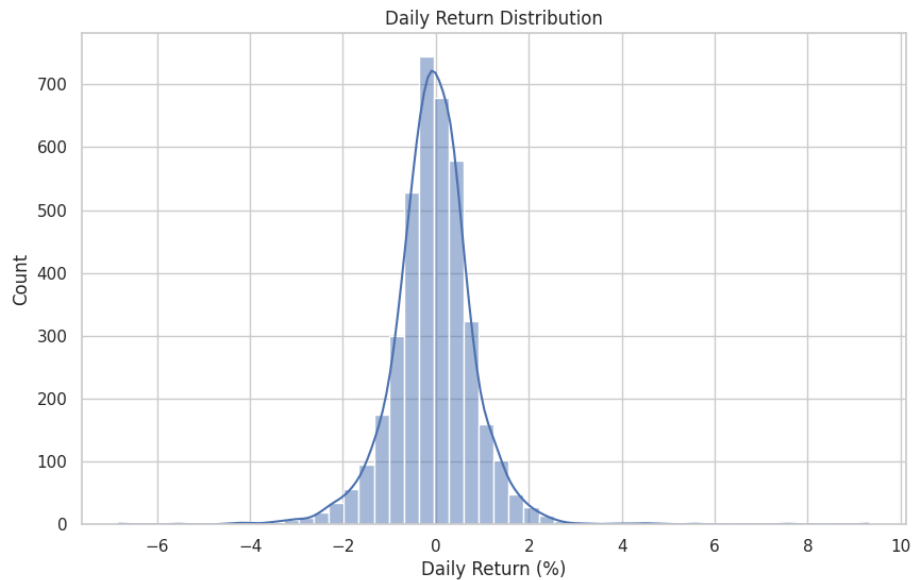


Figure 3: Daily Return Distribution (%) — normal-shaped core with fat tails on both sides

5. Volatility & Risk Analysis

20-day rolling volatility (standard deviation of daily returns) was used as the primary risk gauge. Across the full sample it averages just 0.8, but this average masks sharply episodic behaviour — volatility is calm for years at a time and then spikes abruptly during periods of market stress.

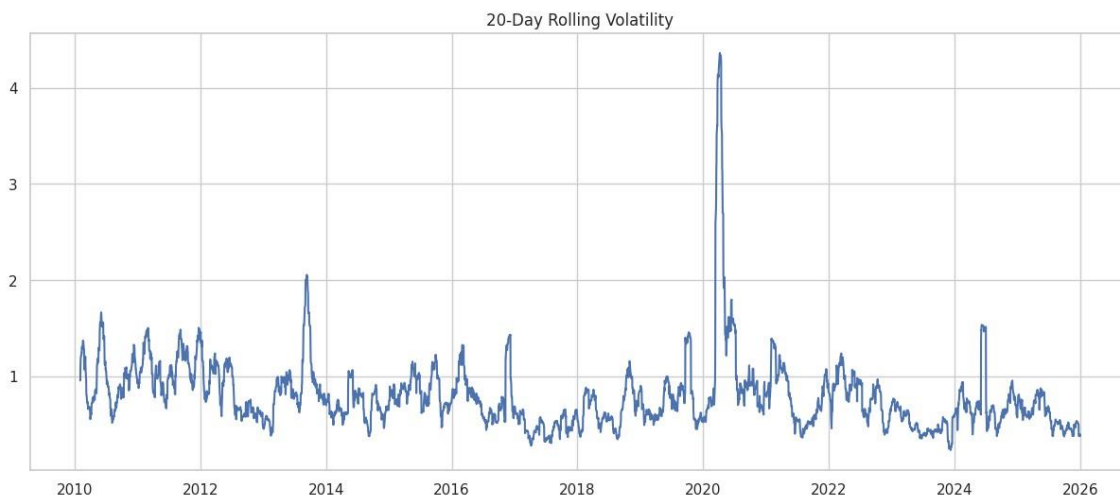


Figure 4: 20-Day Rolling Volatility, 2010–2025 — note the dominant spike above 4.3 in early 2020

• The 2020 Volatility Spike Dwarfs All Other Risk Events

Rolling volatility peaked above 4.3 during the COVID-19 crash — more than 5x the long-run average of 0.8. The next-largest spikes (~2.0 in late 2013, ~1.5 in mid-2024) are an order of magnitude smaller, confirming 2020 as a true outlier event rather than part of a recurring pattern.

Daily trading range (High – Low) tells the same story: the average range is just 129.8 points, but two isolated spikes — one in mid-2020 (~1,610 points) and a larger one in mid-2024 (~1,900 points, coinciding with the 4 June 2024 general-election result session) — stand far above the typical band.

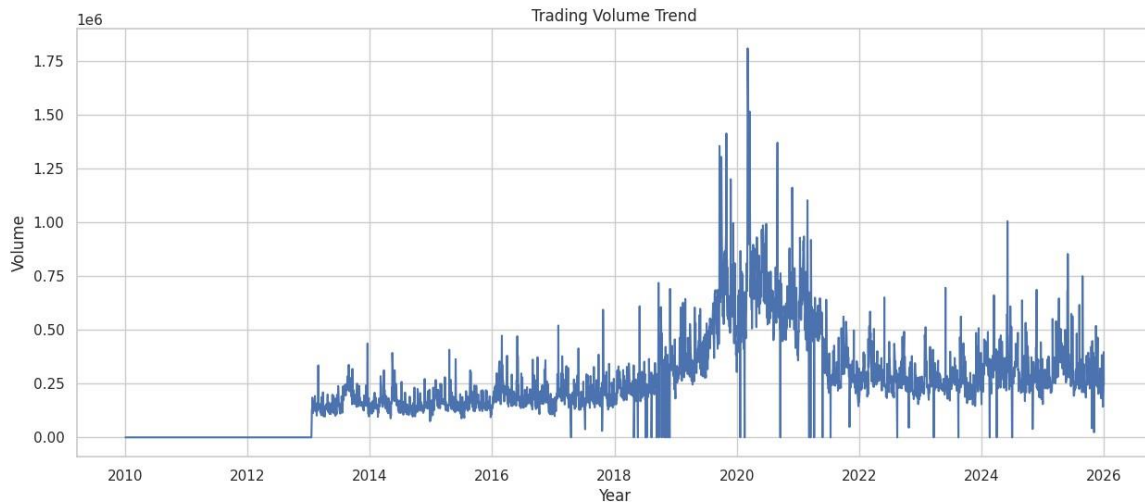


Figure 5: Trading Volume Trend — volume trended structurally higher from 2018 onward and spiked sharply during the 2020 crash

6. Extreme Event Deep-Dive — The 2020 Shock

The ten largest single-day gains and ten largest single-day losses were ranked to identify where tail risk is concentrated. The COVID-19 crash period (February–June 2020) was selected for deep-dive analysis as it accounts for the single largest cluster of extreme-move days in the entire 16-year history.

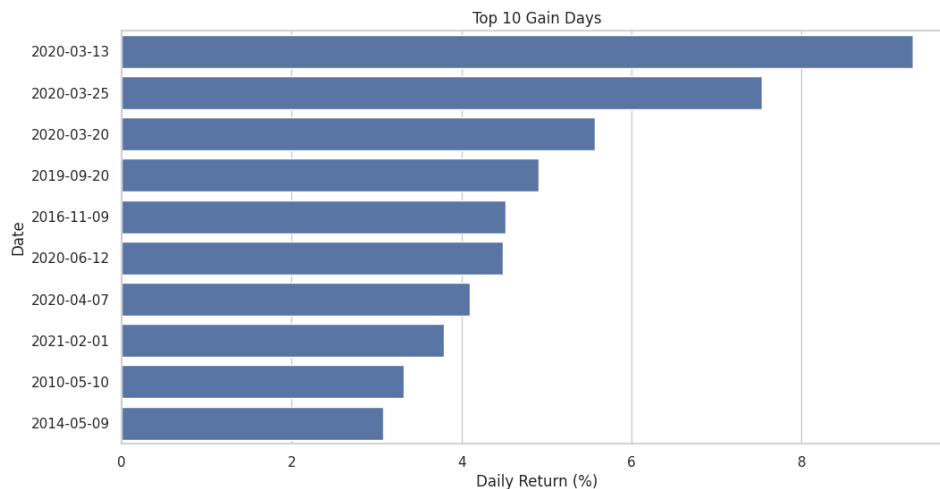


Figure 6: Top 10 Single-Day Gains — ranked by Daily Return (%)

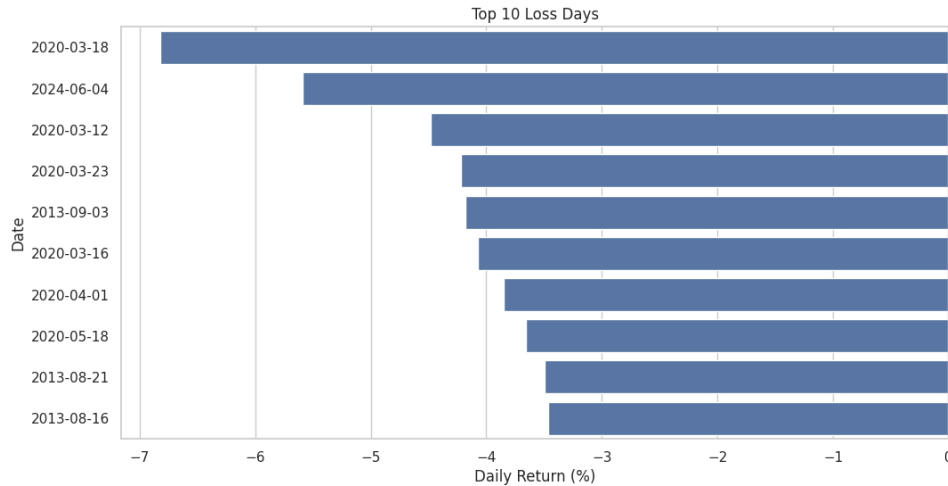


Figure 7: Top 10 Single-Day Losses — ranked by Daily Return (%)

- 6 of the Top 20 Extreme-Move Days Fall Within a Single 4-Month Window**

Of the 10 biggest gains and 10 biggest losses, six (13, 18, 20, 23, 25 March 2020 and 7 April 2020, plus 12 March 2020 and 12 June 2020) occurred during the COVID-19 crash and initial recovery. This is not a randomly distributed tail — it is a single, identifiable macro shock.

- The Second-Worst Day on Record Is a Single Calendar Event, Not a Crash**

4 June 2024 — the day Indian general-election results were declared — produced the second-largest single-day loss in 16 years (−5.6% to −5.7%), alongside the second-largest daily trading-range spike (~1,900 points). This confirms scheduled political/macro event days carry materially elevated tail risk even outside of systemic crises.

Outside the COVID and election-result clusters, the remaining extreme days (August 2013, September 2013, May 2010, May 2014, September 2019, November 2016, February 2021) are isolated single-day events spread across more than a decade, with no further clustering — consistent with idiosyncratic news-driven moves rather than a recurring structural pattern.

7. Trend Analysis — Moving Averages & the Cumulative Return Anomaly

20-day and 50-day moving averages track the closing price almost perfectly (correlation ≈ 1.00 with Close), confirming they add limited standalone diversification value as a signal but are useful for confirming established trend direction at a glance.

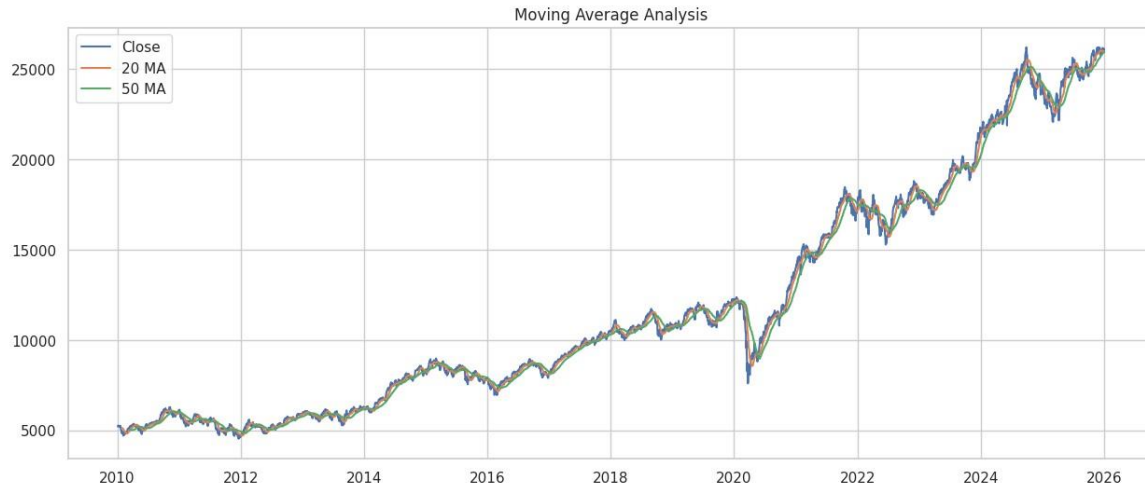


Figure 8: Close Price vs. 20-Day and 50-Day Moving Averages

The most important and counter-intuitive finding in this analysis emerges from the Cumulative Return series, which compounds the daily open-to-close return $((\text{Close} - \text{Open}) / \text{Open})$ starting from a base of 1.0.



Figure 9: Cumulative Return from Daily Open→Close Returns — declining steadily from 1.0 to ~0.07 even as the Close price itself rose more than 5x

• The Cumulative Return Anomaly: Intraday Trading Has Lost Money for 16 Years

Despite the Close price rising from ~₹5,200 to ₹26,216, the compounded open-to-close return series falls almost monotonically from 1.0 to roughly 0.07 over the same period. This means a strategy that bought at the open and sold at the close, every single day for 16 years, would have lost the vast majority of its capital — even though buy-and-hold investors made over 5x their money. The entire divergence is explained by the overnight/opening Gap (Open – Previous Close): nearly all of NIFTY 50's long-run appreciation happens between one day's close and the next day's open, not during the trading session itself. This is corroborated by Section 3's findings that Loss Days (53.0%) outnumber Profit Days (47.0%) and that the average open-to-close return is negative (–0.06%).

This is the single most actionable structural insight in the dataset: it implies that pure intraday (open-to-close) trading strategies on the NIFTY 50 have faced a persistent structural headwind over the full

sample period, while buy-and-hold or overnight-carry exposure has captured essentially all of the index's long-term return.

8. Time-Based & Seasonal Patterns

Three calendar dimensions were examined: month, quarter, and day of week.

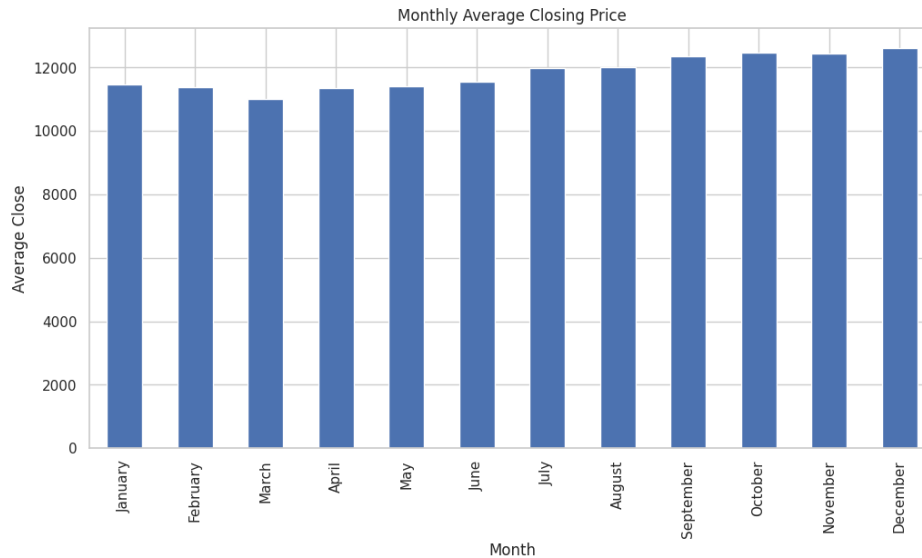


Figure 10: Monthly Average Closing Price

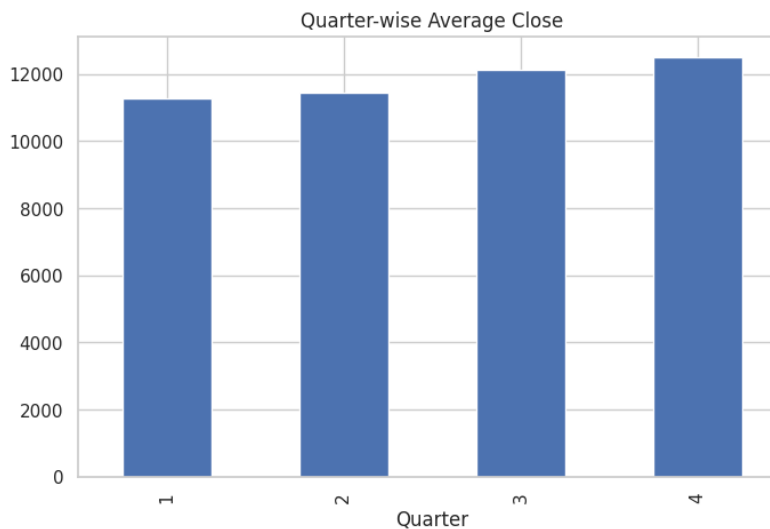


Figure 11: Quarter-wise Average Closing Price

- Month & Quarter “Seasonality” Reflects the Long-Term Uptrend, Not a Repeating Calendar Effect**

Average closing price rises steadily from January through December (and from Q1 through Q4). This pattern looks seasonal but is better explained by the index's persistent multi-year uptrend interacting with how trading days are distributed across the calendar — later months/quarters in

any multi-year window simply average in more of the higher, later-year prices. This should not be read as evidence of a repeating month-of-year trading edge.

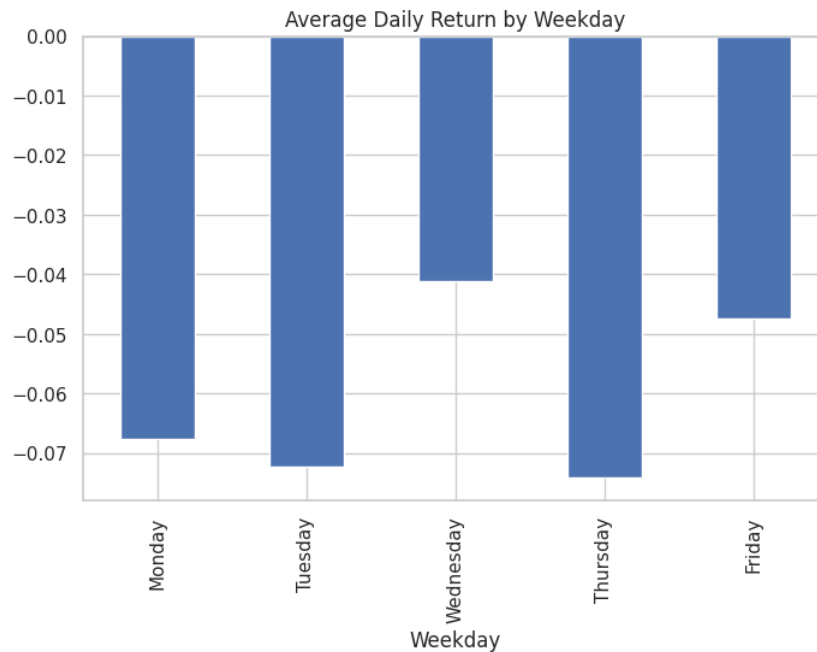


Figure 12: Average Daily Return (Open→Close) by Weekday

- **Every Weekday Shows a Negative Average Open→Close Return**

Unlike the month/quarter pattern, the weekday effect is a genuine, return-based finding: all five trading weekdays average a negative open-to-close return. Thursday is weakest ($\sim -0.074\%$) and Wednesday least negative ($\sim -0.041\%$). This reinforces the Section 7 finding — negative intraday returns are pervasive across every weekday, not concentrated in any one day.

9. Correlation Analysis

A full correlation matrix was computed across all numeric features to understand how price, volume, and derived metrics relate to one another.

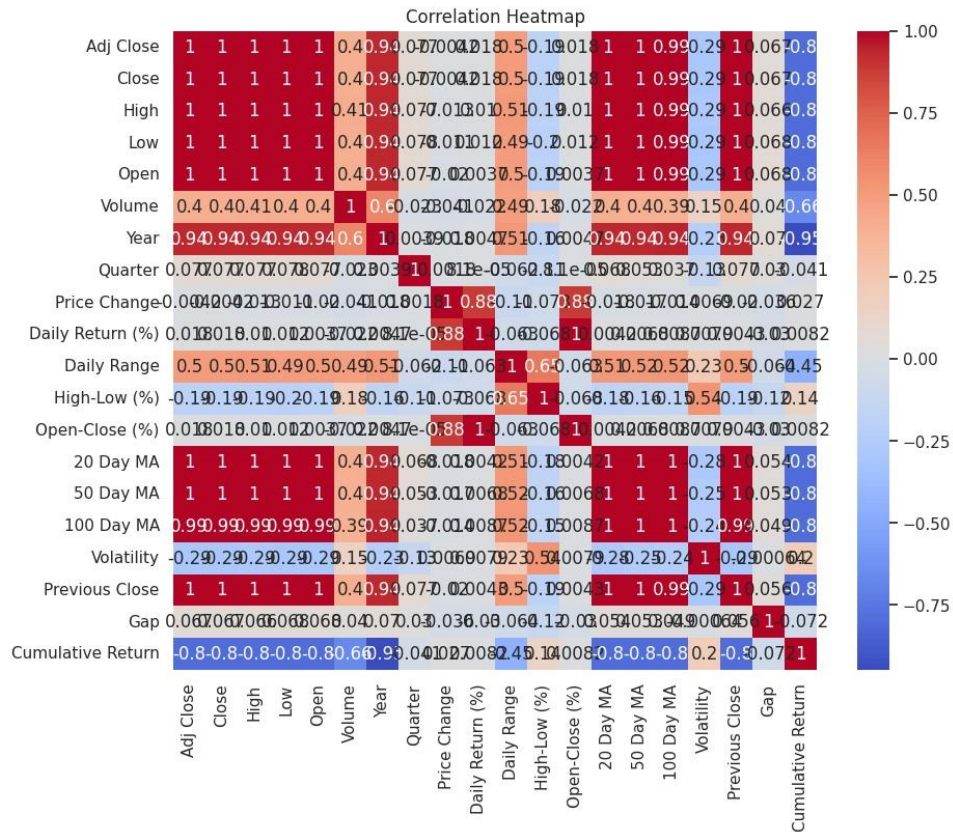


Figure 13: Correlation Heatmap Across All Engineered Features

- Open, High, Low, Close, Adj Close, and the moving averages are all almost perfectly correlated (≈ 1.00) with each other — as expected for OHLC price series of the same instrument.
- Year correlates strongly with price level (≈ 0.94), quantifying the strength of the long-term uptrend.
- Cumulative Return correlates strongly negatively with price level and Year (≈ -0.8), numerically confirming the Section 7 anomaly: as the index price rose over time, the compounded open-to-close return series fell.
- Volume correlates moderately with price level (≈ 0.40 – 0.41) and only weakly with Volatility (≈ 0.15), suggesting volume growth over the sample is driven more by structural market development (more participants, more liquidity over 16 years) than by short-term risk spikes alone.
- Gap (Open – Previous Close) shows only a weak direct correlation with price level (≈ 0.06 – 0.07) despite being the structural driver of long-run returns identified in Section 7 — its effect operates through compounding many small values over thousands of days, not through a strong single-period linear relationship.

10. Predictive Modeling Outlook

This phase of the project focused on exploratory data analysis, feature engineering, and business-insight generation; no predictive model was trained on this dataset. Given the strength of the

patterns surfaced above, three modeling directions are recommended as a natural next phase, mirroring the predictive-modeling stage of comparable analytics projects:

- Volatility forecasting (e.g., GARCH or an ML regression on lagged volatility/range features) to provide forward-looking risk estimates rather than only the backward-looking 20-day rolling measure used here.
- Next-day direction classification (Up/Down) using lagged returns, volume, and volatility as features — directly testing whether the negative average open-to-close return is exploitable or merely a structural bias with no actionable signal.
- Event-flag modeling that explicitly tags scheduled macro/political event dates (budget day, election results, central bank decisions) as a feature, given Section 6's finding that such days carry disproportionate tail risk.

11. Strategic Recommendations

CRITICAL

1. Re-Examine Any Intraday (Open-to-Close) Trading Strategy on This Index

The data shows a persistent negative average open-to-close return (−0.06%) with more Loss Days than Profit Days across 16 years. Any strategy that buys at the open and exits at the close should be stress-tested against this structural headwind before live deployment. Strategies that capture the overnight Gap (hold from close to next open) merit comparative evaluation, since this is where the index's actual long-run return appears to be concentrated.

HIGH

2. Build a Volatility-Regime Risk Management Framework

20-day rolling volatility provided clear early warning of the 2020 shock, rising well above its 0.8 baseline before peaking above 4.3. A volatility-regime trigger (e.g., position-size reduction or hedge initiation once rolling volatility crosses 1.5–2x its trailing average) would have materially reduced drawdown risk during the largest stress event in the sample.

HIGH

3. Treat Scheduled Macro/Political Event Days as a Distinct Risk Category

4 June 2024 (election results) produced the second-worst single-day loss and the second-largest trading-range spike in 16 years, entirely outside of any broader crisis. Maintain a calendar of known high-impact event dates (budget day, election results, major central-bank decisions) and apply tighter risk limits or hedges heading into them.

MEDIUM

4. Investigate the Thursday Weakness in Open-to-Close Returns

Thursday shows the weakest average open-to-close return (−0.074%) of any weekday, with Wednesday the least negative (−0.041%). This warrants out-of-sample validation before being treated as a tradable edge, but is worth a dedicated follow-up study given the consistency across the full 16-year sample.

MEDIUM

5. Use Moving Averages for Trend Confirmation, Not Standalone Signals

20-day and 50-day moving averages move almost in lockstep with Close (correlation ≈ 1.00), so they add limited value as an independent signal. Pair them with the volatility-regime framework above (Recommendation 2) for more differentiated entry/exit logic.

LOW

6. Extend the Dataset to a Dividend-/Total-Return-Adjusted Series

The Cumulative Return metric in this analysis is built purely from price (Open/Close), and does not capture dividends or corporate actions. Cross-checking the Section 7 anomaly against an official Total Return Index would clarify how much of the gap is a pure price-return phenomenon versus an artifact of excluding income return.

LOW

7. Build Constituent-Level Dashboards as a Follow-On Study

This analysis is index-level only. A follow-on study decomposing index moves into individual constituent contributions would clarify whether the overnight-gap effect and Thursday weakness are broad-based across the 50 constituent stocks or concentrated in a subset.

12. Conclusion

This analysis has surfaced a clear and important picture: the NIFTY 50 has delivered strong long-term, buy-and-hold price appreciation — more than 5x over 16 years — while simultaneously exhibiting a persistent negative average return within the trading session itself. Both facts are true at once, and reconciling them (Section 7's cumulative return anomaly) is the most actionable finding in this report for anyone designing a trading or investment strategy on this index.

Risk is highly episodic rather than evenly spread: the 2020 COVID-19 crash dominates the volatility and extreme-move statistics, while the 4 June 2024 election-result session demonstrates that scheduled macro/political events can independently produce crisis-level single-day moves. Calendar effects beyond the weekday-level open-to-close pattern (Section 8) should be interpreted cautiously, as month/quarter patterns largely reflect the underlying multi-year uptrend rather than a repeating seasonal edge.

Priority Area	Current State	Recommended Action
Intraday strategy risk	Avg. open→close return -0.06% ; 53% Loss Days	Stress-test/avoid pure intraday strategies; evaluate overnight-gap capture
Volatility regime risk	Spiked $>5x$ baseline in 2020	Deploy volatility-triggered position sizing / hedging
Event-day risk	Election result day = 2nd-worst day on record	Maintain event calendar; tighten risk limits pre-event
Weekday effect	Thursday weakest, Wednesday strongest (least negative)	Validate out-of-sample before acting
Return data scope	Price-only cumulative return	Cross-check against Total Return Index incl. dividends

With focused attention on the structural intraday-return anomaly and a disciplined approach to volatility- and event-driven risk, investors and risk managers can build a more complete picture of NIFTY 50 behaviour than the headline price chart alone provides — one that explains not just where

the index has been, but why its long-term return and its day-to-day trading behaviour tell two different stories.